McCann model for rhomohedral graphene

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1 Introduction

This report analyzes the behavior of ABC-stacked multilayer graphene using the McCann model. We specifically compare the cases for $N = 1$ (monolayer) and $N = 5$ layers, focusing on their energy bands, Berry curvature, and the convergence of the local Chern number contribution.

2 Model and Analytical Equations

The system is described by a 2x2 effective Hamiltonian in the vicinity of a K point (low energy):

$$H = \begin{pmatrix} \Delta & X(\mathbf{k}) \\ X^\dagger(\mathbf{k}) & -\Delta \end{pmatrix} \tag{1}$$

where 2Δ is the energy gap and $X(\mathbf{k})$ is the off-diagonal coupling term. In polar coordinates (k, ϕ), the term X for N layers is given by:

$$X(k, \phi) = \frac{(vke^{i\xi\phi})^N}{(-\gamma_1)^{N-1}} \tag{2}$$

where $v = \frac{\sqrt{3}}{2}\gamma_0$ in natural units ($a = 1$, $\hbar = 1$) and $\xi = \pm 1$ is the valley index.

2.1 Energy Bands

The eigenvalues of the Hamiltonian represent the energy bands:

$$\varepsilon(\mathbf{k}) = \pm E(\mathbf{k}) = \pm\sqrt{|X(\mathbf{k})|^2 + \Delta^2} \tag{3}$$

2.2 Eigenstates

The normalized eigenstates for the positive and negative bands are given by:

$$|\psi_+\rangle = \frac{1}{\mathcal{N}_+}\begin{pmatrix} E + \Delta \\ X^\dagger \end{pmatrix}, \quad |\psi_-\rangle = \frac{1}{\mathcal{N}_-}\begin{pmatrix} -X \\ E + \Delta \end{pmatrix} \tag{4}$$

where $\mathcal{N}_\pm$ are the normalization constants.

2.3 Berry Curvature

The Berry curvature Ω is calculated using the Kubo-like formula. For the results presented in this report, we specifically show the Berry curvature calculated for the positive energy band:

$$\Omega_n = i \sum_{m \neq n} \frac{\langle\psi_n|\nabla_{\mathbf{k}} H|\psi_m\rangle \times \langle\psi_m|\nabla_{\mathbf{k}} H|\psi_n\rangle}{(E_n - E_m)^2} \tag{5}$$

Here, the gradient is calculated in polar coordinates as:

$$\nabla_{\mathbf{k}} = \frac{\partial}{\partial k}\vec{e}_k + \frac{1}{k}\frac{\partial}{\partial \phi}\vec{e}_\phi \tag{6}$$

3 Numerical Results

3.1 Energy Bands Comparison

Figure 1 shows the energy bands for $N = 1$ and $N = 5$. For $N = 1$, the dispersion is linear at high k (Dirac-like), while for $N = 5$ it is much flatter near $k = 0$ due to the k^N dependence.

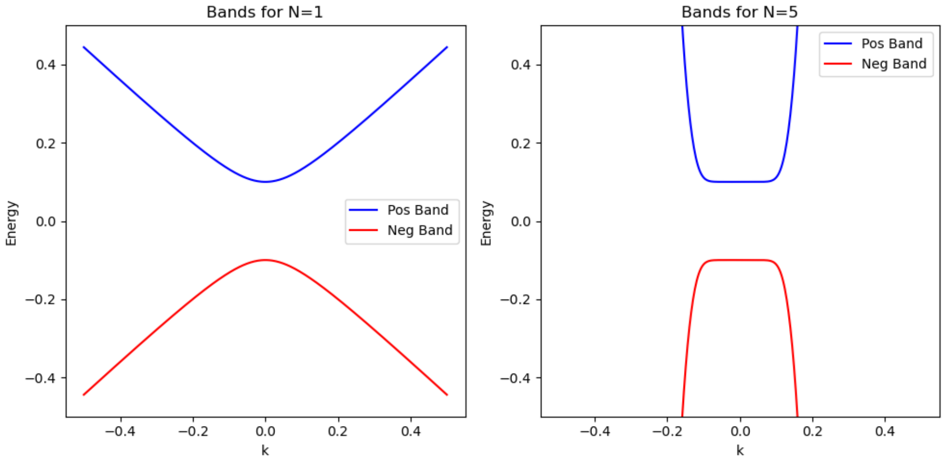

Figure 1: Energy bands for N=1 and N=5 layers.

3.2 Berry Curvature

The Berry curvature for $N = 1$ peaks sharply at $k = 0$. However, for $N = 5$, the curvature is suppressed at the center and peaks at finite k (Figure 2).

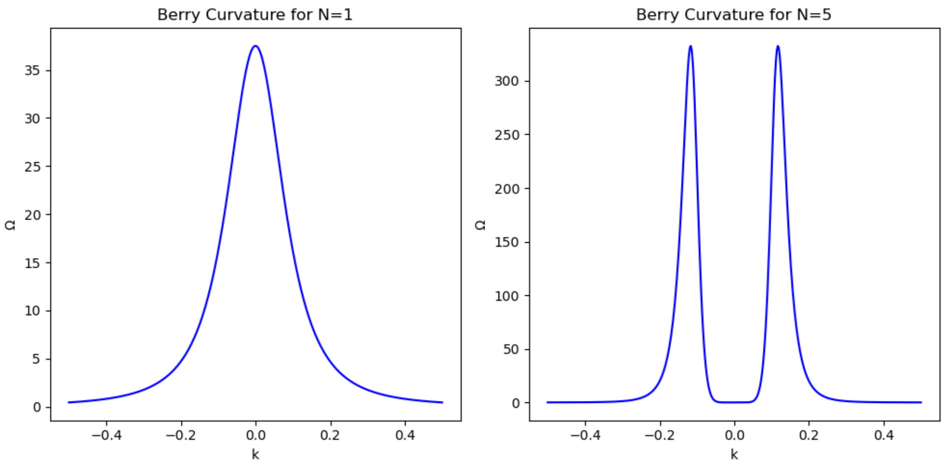

Figure 2: Berry curvature comparison (Positive Band).

4 Berry Curvature Integration

The local Chern number contribution is obtained by integrating the Berry curvature over a region in k-space. While the McCann model is isotropic and naturally expressed in polar coordinates, the numerical integration is performed over a uniform Cartesian meshgrid (k_x, k_y). This choice provides several numerical advantages:

- **Uniform Sampling Density:** In a Cartesian grid, every point represents an identical area element $\Delta A = \Delta k_x \Delta k_y$. In a polar grid, the area element $dA = k\,dk\,d\phi$ is proportional to k and vanishes as $k \to 0$. Since the Berry curvature is often highly concentrated near the Dirac point ($k = 0$ for monolayer graphene), the Cartesian grid ensures high sampling density where it is most needed without requiring a non-linear radial distribution.

- **Singularity Avoidance:** The Cartesian coordinate system avoids the coordinate singularity at the origin inherent to polar grids, which simplifies the handling of the peak in Ω at $k = 0$ for $N = 1$.

To ensure a consistent comparison between different layer numbers N, we define the integration domain using an energy cutoff E_{max} rather than a fixed momentum radius. The integral is evaluated numerically as:

$$C_{local}(E_{max}) = \frac{1}{2\pi} \iint_{E(\mathbf{k}) \le E_{max}} \Omega_{pos}(\mathbf{k}) dk_x dk_y \approx \frac{\Delta A}{2\pi} \sum_{i \in \text{mask}} \Omega_{pos}(\mathbf{k}_i) \tag{7}$$

where the sum runs over all points k_i satisfying the energy condition $E(\mathbf{k}_i) \le E_{max}$, and ΔA is the constant area per grid point.

Figure 3 shows that as the energy cutoff E_{max} increases, the integral converges to $N/2$.

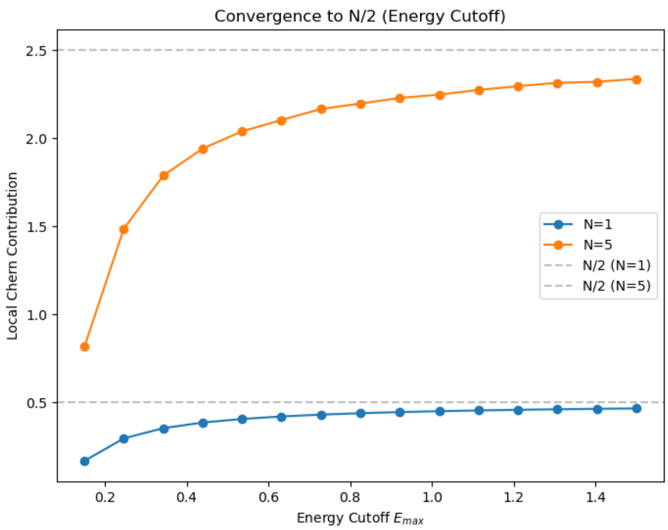

Figure 3: Convergence of the local Chern number contribution to $N/2$ as a function of the energy cutoff E_{max}.

5 Valley Comparison

The two K-valleys in the hexagonal lattice, indexed by $\xi = \pm 1$, are related by time-reversal symmetry. While the energy dispersion is identical for both valleys, the Berry curvature of a given band changes its sign:

$$\begin{aligned} \mathcal{T}:\ & \mathbf{k} \to -\mathbf{k} \\ & E(-\mathbf{k}) = E(\mathbf{k}) \\ & \Omega(-\mathbf{k}) = -\Omega(\mathbf{k}) \end{aligned} \tag{8}$$

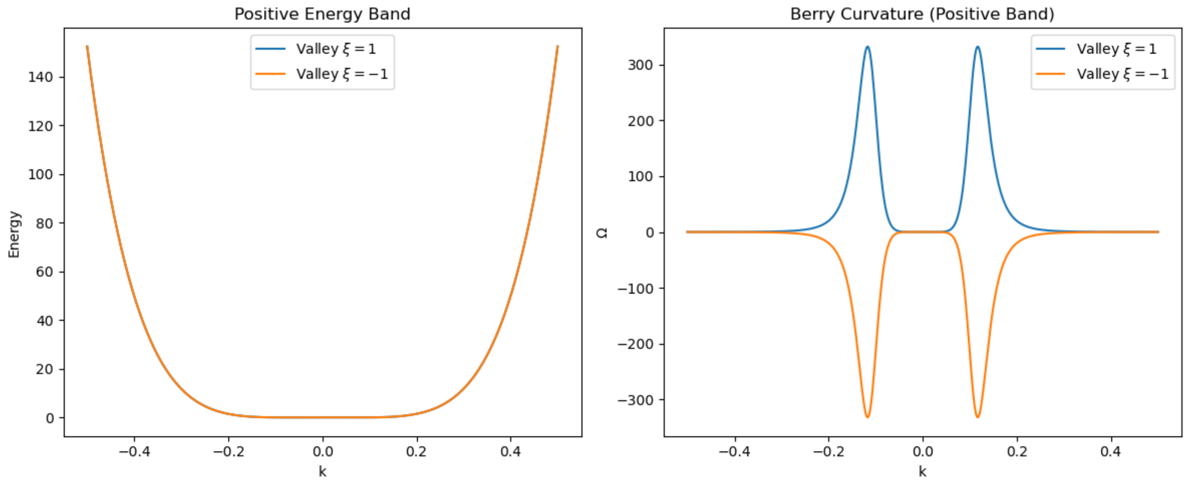

Figure 4: Comparison of the positive energy band (left) and Berry curvature (right) for valleys $\xi = 1$ and $\xi = -1$ ($N = 5$ layers).

Figure 4 compares the positive energy band and the corresponding Berry curvature for $N = 5$ layers, showing the sign inversion between the two K valleys.

Consequently, the local Chern number contributions for $N = 5$ have opposite signs. The integration of the Berry curvature over a circle of radius k=1 gives:

- Valley $\xi = 1$: $C_{local} = 2.499949$
- Valley $\xi = -1$: $C_{local} = -2.499949$

The total Chern number for the system, considering both valleys, would vanish in the presence of time-reversal symmetry ($C_{tot} = C_K + C_{K'} = 0$).

6 Conclusion

The N-layer McCann system exhibits a topological Berry phase of $N\pi$, leading to a local Chern contribution of $N/2$ per valley in the gapped state. The distribution of the Berry curvature becomes increasingly non-trivial as N increases, with the peak shifting away from the Dirac point while the total topological charge remains conserved and proportional to N.